Entity-Relationship (ER) Overview

- **theaters**: Stores multiplex and cinema location details.
- **screens**: Defines individual auditoriums or screens inside a theater.
- **movies**: Contains film metadata, duration, and language.
- **shows**: Represents a specific screening of a movie in a screen at a specific time.
- **seats**: Defines the physical seat grid and layout belonging to a screen.
- **show_seats**: Maps seats to specific shows, tracking real-time status (AVAILABLE, LOCKED, BOOKED) with a version column for optimistic concurrency control.
- **bookings**: Acts as the transaction header tracking user booking and payment status.
- **booking_items**: Line items securely linking a booking to specific show_seats.

Table Attributes & Example Rows

1. theaters

• Attributes: theater_id (PK, INT), name (VARCHAR), city (VARCHAR), address (TEXT), created_at (TIMESTAMP)

• Example Row: (1, 'PVR Cinemas: Phoenix Mills', 'Mumbai', 'Lower Parel, Mumbai', '2026-09-30 21:07:00')

2. screens

• Attributes: screen_id (PK, INT), theater_id (FK, INT), name (VARCHAR), total_seats (INT)

• Example Row: (1, 1, 'Audi 1 - IMAX', 100)

3. movies

• Attributes: movie_id (PK, INT), title (VARCHAR), duration_minutes (INT), language (VARCHAR), release_date (DATE)

• Example Row: (1, 'Interstellar (Re-release)', 169, 'English', '2026-09-01')

4. shows

• Attributes: show_id (PK, INT), movie_id (FK, INT), screen_id (FK, INT), start_time (DATETIME), end_time (DATETIME), base_price (DECIMAL)

• Example Row: (1, 1, 1, '2026-10-05 18:30:00', '2026-10-05 21:20:00', 450.00)

5. seats

• Attributes: seat_id (PK, INT), screen_id (FK, INT), row_number (VARCHAR), seat_number (INT), seat_type (ENUM)

• Example Row: (1, 1, 'A', 1, 'PREMIUM')

6. show_seats

• Attributes: show_seat_id (PK, INT), show_id (FK, INT), seat_id (FK, INT), status (ENUM), version (INT)

• Example Row: (1, 1, 1, 'AVAILABLE', 0)

7. bookings

• Attributes: booking_id (PK, VARCHAR UUID), user_id (INT), show_id (FK, INT), total_amount (DECIMAL), payment_status (ENUM), created_at (TIMESTAMP)

• Example Row: ('a1b2c3d4-e5f6...', 101, 1, 900.00, 'PENDING', '2026-09-30 22:00:00')

8. booking_items

• Attributes: booking_item_id (PK, INT), booking_id (FK, VARCHAR), show_seat_id (FK, INT), price (DECIMAL)

• Example Row: (1, 'a1b2c3d4-e5f6...', 1, 450.00)

Task P1 SQL Schema (MySQL DDL)

```
CREATE DATABASE IF NOT EXISTS bookmyshow_db;
USE bookmyshow_db;

CREATE TABLE theaters (
   theater_id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(150) NOT NULL,
   city VARCHAR(100) NOT NULL,
   address TEXT NOT NULL,
   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB;

CREATE TABLE screens (
   screen_id INT AUTO_INCREMENT PRIMARY KEY,
   theater_id INT NOT NULL,
   name VARCHAR(50) NOT NULL,
   total_seats INT NOT NULL,
   FOREIGN KEY (theater_id) REFERENCES theaters(theater_id) ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE TABLE movies (
   movie_id INT AUTO_INCREMENT PRIMARY KEY,
   title VARCHAR(200) NOT NULL,
   duration_minutes INT NOT NULL,
   language VARCHAR(50) NOT NULL,
   release_date DATE NOT NULL
) ENGINE=InnoDB;
```

```
CREATE TABLE shows (
    show_id INT AUTO_INCREMENT PRIMARY KEY,
    movie_id INT NOT NULL,
    screen_id INT NOT NULL,
    start_time DATETIME NOT NULL,
    end_time DATETIME NOT NULL,
    base_price DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (movie_id) REFERENCES movies(movie_id) ON DELETE CASCADE,
    FOREIGN KEY (screen_id) REFERENCES screens(screen_id) ON DELETE CASCADE,
    INDEX idx_theater_date (screen_id, start_time)
) ENGINE=InnoDB;

CREATE TABLE seats (
    seat_id INT AUTO_INCREMENT PRIMARY KEY,
    screen_id INT NOT NULL,
    row_number VARCHAR(5) NOT NULL,
    seat_number INT NOT NULL,
    seat_type ENUM('REGULAR', 'PREMIUM', 'VIP') NOT NULL DEFAULT 'REGULAR',
    FOREIGN KEY (screen_id) REFERENCES screens(screen_id) ON DELETE CASCADE,
    UNIQUE KEY uk_screen_seat (screen_id, row_number, seat_number)
) ENGINE=InnoDB;

CREATE TABLE show_seats (
    show_seat_id INT AUTO_INCREMENT PRIMARY KEY,
    show_id INT NOT NULL,
    seat_id INT NOT NULL,
    status ENUM('AVAILABLE', 'LOCKED', 'BOOKED') NOT NULL DEFAULT 'AVAILABLE',
    version INT NOT NULL DEFAULT 0,
    FOREIGN KEY (show_id) REFERENCES shows(show_id) ON DELETE CASCADE,
```

```
    FOREIGN KEY (seat_id) REFERENCES seats(seat_id) ON DELETE CASCADE,
    UNIQUE KEY uk_show_seat (show_id, seat_id)
) ENGINE=InnoDB;

CREATE TABLE bookings (
    booking_id VARCHAR(36) PRIMARY KEY,
    user_id INT NOT NULL,
    show_id INT NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    payment_status ENUM('PENDING', 'SUCCESS', 'FAILED', 'REFUNDED') NOT NULL DEFAULT 'PENDING',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (show_id) REFERENCES shows(show_id)
) ENGINE=InnoDB;

CREATE TABLE booking_items (
    booking_item_id INT AUTO_INCREMENT PRIMARY KEY,
    booking_id VARCHAR(36) NOT NULL,
    show_seat_id INT NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE,
    FOREIGN KEY (show_seat_id) REFERENCES show_seats(show_seat_id),
    UNIQUE KEY uk_show_seat_booked (show_seat_id)
) ENGINE=InnoDB;
```

Task P2 Show Listing Query

```
SELECT
    m.movie_id,
    m.title AS movie_title,
    m.duration_minutes,
    m.language,
    s.show_id,
    sc.name AS screen_name,
    s.start_time,
    s.end_time,
    s.base_price
FROM theaters t
JOIN screens sc ON t.theater_id = sc.theater_id
JOIN shows s ON sc.screen_id = s.screen_id
JOIN movies m ON s.movie_id = m.movie_id
WHERE t.theater_id = 1
  AND DATE(s.start_time) = '2026-10-05'
ORDER BY m.title, s.start_time;
```